

Stellar Interiors, Nucleosynthesis & Evolution

Problem Set

fizicyst

This problem set accompanies the *Stellar Interiors, Nucleosynthesis & Evolution* handout. Section A has five objective (multiple-choice) questions; Section B has three subjective questions, pitched a little harder than Section A. Use $G = 6.674 \times 10^{-11} \text{ N m}^2\text{kg}^{-2}$, $\hbar = 1.055 \times 10^{-34} \text{ J s}$, $c = 3 \times 10^8 \text{ m/s}$, $k = 1.381 \times 10^{-23} \text{ J/K}$, $m_H = 1.673 \times 10^{-27} \text{ kg}$, $M_\odot = 1.989 \times 10^{30} \text{ kg}$, and $1 \text{ pc} = 3.086 \times 10^{16} \text{ m}$ where needed, unless a question tells you otherwise.

Section A — Objective

O1. A star has mass $M = 0.5 M_\odot$. Using $\tau_{\text{MS}} \propto M^{-2.5}$, its main-sequence lifetime relative to the Sun's is closest to:

- (a) $0.5\times$
- (b) $2\times$
- (c) $5.7\times$
- (d) $11.3\times$

O2. Iron-56 cannot release energy via further fusion or fission because:

- (a) it has the *lowest* binding energy per nucleon of any isotope
- (b) it sits at the *peak* of the binding-energy-per-nucleon curve
- (c) it is radioactive and decays into lighter elements
- (d) it has too few protons to fuse with anything else

O3. A black hole has mass $M = 3 M_\odot$. Taking the Schwarzschild radius for one solar mass to be $R_s \approx 3 \text{ km}$, this black hole's Schwarzschild radius is closest to:

- (a) 3 km
- (b) 9 km
- (c) 1 km
- (d) 27 km

O4. Which one of the following statements is **false**?

- (a) Electron degeneracy pressure depends only on density, not temperature.
- (b) Above the Chandrasekhar limit, no stable white dwarf configuration exists.
- (c) Neutron stars can support more mass than white dwarfs because neutrons have a *smaller* Fermi momentum than electrons at the same density.
- (d) The Tolman–Oppenheimer–Volkoff limit for neutron stars is roughly $2\text{--}3 M_\odot$.

O5. If a star’s core temperature increases by a factor of 2, the approximate factor by which its CNO-cycle energy generation rate increases (taking $\varepsilon_{\text{CNO}} \propto T^{18}$) is closest to:

- (a) 16
- (b) 36
- (c) 8
- (d) 2.6×10^5

Section B — Subjective

S1. A protostellar cloud of mass $M = 1 M_\odot$ collapses from an initial radius $R_i = 1$ pc (where its gravitational PE is negligible) to $R_f = 0.01$ pc. Model the cloud as a uniform sphere, so $U = -\frac{3GM^2}{5R}$.

- (i) Find the magnitude of the gravitational PE released, $|\Delta U|$, in the collapse.
- (ii) Using the virial theorem argument of Section 4.2, find how much of this released energy goes into heating the gas (ΔK), and how much is radiated away.
- (iii) If this thermal energy is shared among $N \approx 1.2 \times 10^{57}$ hydrogen atoms (roughly $1 M_\odot$ worth), estimate the mean thermal energy per particle and hence a rough temperature, using $\frac{3}{2}kT$ per particle. Comment on whether this is a plausible temperature for a molecular cloud at this stage.

S2. Relativistic degenerate electron pressure scales as $P_{\text{deg,rel}} \sim \hbar c n^{4/3}$, where n is the electron number density. For a star of mass M and radius R made of hydrogen (so $n \sim M/(m_H R^3)$, roughly one electron per baryon), and using the order-of-magnitude hydrostatic estimate $P_{\text{grav}} \sim GM^2/R^4$:

- (i) Substitute $n \sim M/(m_H R^3)$ into $P_{\text{deg,rel}}$, and set $P_{\text{deg,rel}} \sim P_{\text{grav}}$.
- (ii) Show that R cancels out of this balance condition entirely, and solve for the critical mass M_{crit} purely in terms of $\hbar$, c , G , and m_H .
- (iii) Evaluate M_{crit} numerically, in both kg and solar masses, and compare your order-of-magnitude estimate (which drops all numerical prefactors) to the actual Chandrasekhar limit of $1.4 M_\odot$.

S3. A star has mass $M = 3 M_{\odot}$ and is dominated by the CNO cycle.

- (i) Using $\tau_{\text{MS}} \propto M^{-2.5}$ and $\tau_{\odot} \approx 10$ Gyr, estimate this star's main-sequence lifetime.
- (ii) Using $\varepsilon_{\text{CNO}} \propto T^{17}$, find the factor by which the CNO energy generation rate increases for just a 15% increase in core temperature.
- (iii) Given the extreme sensitivity you just found in (ii), briefly explain why the CNO cycle effectively acts as a “thermostat” for massive stars — i.e. why their core temperatures end up clustering in a fairly narrow range across a wide range of masses, even though luminosity (and hence lifetime, from (i)) varies by orders of magnitude.